BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Truth Tables and Combined Logic

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. How can you recognise a NOT gate symbol?

2. What does an encryption utility produce?
   A) A smaller file
   B) Ciphertext
   C) A sorted folder
   D) Extra storage space

3. "Publicly posting all personal records always improves privacy." Correct this. Explain why.

4. A login must get details, check them and choose the next step. "Algorithmic thinking means guessing steps in a random order." Correct this. Why does order matter?

Today's objectives

Explain the core ideas of truth tables and combined logic using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Combined logic
Work out intermediate gate outputs before calculating the final result.

Operator precedence
Use brackets to clarify order when translating logic diagrams.

Three inputs
A truth table has $2^3 = 8$ unique input combinations.

Intermediate columns
Recording a gate's result reduces mistakes in multi-stage circuits.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Combined logic
Work out intermediate gate outputs before calculating the final result.

Operator precedence
Use brackets to clarify order when translating logic diagrams.

Three inputs
A truth table has $2^3 = 8$ unique input combinations.

Intermediate columns
Recording a gate's result reduces mistakes in multi-stage circuits.

I Do - Worked model

Complete Q = (A AND B) OR C for all eight input combinations.

- List 000,001,010,011,100,101,110,111 in ABC order.
- Find X=A AND B: 0,0,0,0,0,0,1,1.
- Then Q=X OR C: 0,1,0,1,0,1,1,1.
- Check rows with C=1 must have Q=1.

We Do - Guided practice

1. Find Q=(A OR B) AND C when A=1, B=0, C=0 and then C=1.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the condition so A with B, or C alone, raises an alert.

```
A = true
B = false
C = true
if ... then
print("Investigate")
endif
```

You Do - Independent exam questions

Q2. How many rows are needed for a 3-input truth table? [2 marks]

Q3. Evaluate (A AND B) OR C when A=1 B=0 C=1. [2 marks]

Q4. Explain the benefit of an intermediate X column. [2 marks]

Exit Ticket - True or False?

A three-input truth table has only four rows.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. It is a one-input inverter, usually a triangle with a small output bubble.

2. B - Ciphertext

3. People may lose control over how identifying information is used or shared.

4. Plan clear steps in the right order to solve a problem. Keep needed details, split the problem into parts, then link the steps.

I Do - Explain the model

List 000,001,010,011,100,101,110,111 in ABC order.

Find X=A AND B: 0,0,0,0,0,0,1,1.

Then Q=X OR C: 0,1,0,1,0,1,1,1.

Check rows with C=1 must have Q=1.

We Do - Model answers

1. Explain why AND is applied last.

Code First - Answer

```
A = true
B = false
C = true
if (A AND B) OR C then
print("Investigate")
endif
```

Expected output: Investigate

You Do - Self-mark

Q2: $2^3=8$.

Q3: 1.

Q4: Allows accurate separate evaluation of one gate and helps trace errors in combined expressions.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.